

Daily Logic Drill #2 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Answers and Methods.

Section A Rapid Recognition

1. Is “ n is divisible by 4” a sufficient condition for “ n is divisible by 2”?

Answer: Yes.

Method: n divisible by 4 $\Rightarrow$ n divisible by 2, since $4 = 2 \times 2$. “Sufficient” means the condition, if true, guarantees the target.

Investigate further: This is exactly Day 1’s original conditional, now read in necessary/sufficient language: “ A is sufficient for B ” means precisely $A \Rightarrow B$.

2. Is “ n is divisible by 4” a necessary condition for “ n is divisible by 2”?

Answer: No.

Method: “Necessary” means n divisible by 2 $\Rightarrow$ n divisible by 4, which is false: $n = 2$ is divisible by 2 but not 4.

Investigate further: “ A necessary for B ” means $B \Rightarrow A$ — the converse direction of A1. Necessary and sufficient point opposite ways along the same implication.

3. Is “ $x = 3$ ” a sufficient condition for “ $x^2 = 9$ ”?

Answer: Yes.

Method: $x = 3 \Rightarrow x^2 = 9$ directly by substitution.

Investigate further: Compare A4: sufficiency and necessity are independent questions about the same pair of statements.

4. Is “ $x = 3$ ” a necessary condition for “ $x^2 = 9$ ”?

Answer: No.

Method: $x^2 = 9 \Rightarrow x = 3$ is false: $x = -3$ also satisfies $x^2 = 9$ (Day 1, A2).

Investigate further: “ $x = 3$ ” is sufficient but not necessary for “ $x^2 = 9$ ” — the first fully classified pair of the day.

5. True or false: “being a rectangle” is a necessary condition for “being a square”.

Answer: True.

Method: Square $\Rightarrow$ rectangle (Day 1, A3), so “rectangle” is required (necessary) whenever “square”

holds.

Investigate further: Necessary conditions are what you can't do without; they needn't guarantee the target on their own.

6. True or false: “being a rectangle” is a sufficient condition for “being a square”.

Answer: False.

Method: A 2×3 rectangle (Day 1, A4) is a counterexample: it's a rectangle but not a square.

Investigate further: “Rectangle” is necessary but not sufficient for “square” — the exact mirror image of A3/A4's pair.

7. Is “ $n = 6$ ” a sufficient condition for “ n is a multiple of 3”?

Answer: Yes.

Method: $n = 6 \Rightarrow n$ is a multiple of 3, since $6 = 3 \times 2$.

Investigate further: A single value can always be sufficient for a broader property it satisfies — the interesting question is always necessity.

8. Is “ $n = 6$ ” a necessary condition for “ n is a multiple of 3”?

Answer: No.

Method: $n = 9$ is a multiple of 3 but $n \neq 6$, so “ n is a multiple of 3” does not require $n = 6$.

Investigate further: A single value is essentially never necessary for a broader property — necessity almost always demands a whole family, not one instance.

9. True or false: “ n is even” is a necessary condition for “ n is divisible by 4”.

Answer: True.

Method: n divisible by 4 $\Rightarrow n$ even, since $4 = 2 \times 2$.

Investigate further: This is the same pair as A1/A2, viewed from the opposite side — “even” is necessary for “div. by 4” exactly because “div. by 4” is sufficient for “even”.

10. True or false: “ n is even” is a sufficient condition for “ n is divisible by 4”.

Answer: False.

Method: $n = 2$ is even but not divisible by 4.

Investigate further: “Even” is necessary but not sufficient for “divisible by 4” — Section B starts combining necessary and sufficient into a single four-way classification.

Section B Manipulation Drills

1. Classify “ $x > 5$ ” as a condition for “ $x > 0$ ”: necessary, sufficient, both, or neither?

Answer: Sufficient but not necessary.

Method: $x > 5 \Rightarrow x > 0$ (sufficient). But $x > 0 \Rightarrow x > 5$ is false, e.g. $x = 1$ (not necessary).

Investigate further: A “wider” threshold statement is always sufficient for a “narrower” one it’s nested inside, and never necessary for it.

2. Classify “ n is divisible by 6” as a condition for “ n is divisible by 2 and by 3”.

Answer: Necessary and sufficient.

Method: Sufficient: $6 \mid n \Rightarrow 2 \mid n$ and $3 \mid n$ directly. Necessary: if $2 \mid n$ and $3 \mid n$, then since $\gcd(2, 3) = 1$, $\text{lcm}(2, 3) = 6 \mid n$. Both directions hold, so it’s an iff (this restates Day 1’s B1 in classification language).

Investigate further: Whenever a statement and its converse are both true (Day 1), the condition is necessary and sufficient — the two ideas are the same fact in different vocabulary.

3. Classify “ $ab = 0$ ” as a condition for “ $a = 0$ ” (where a, b are real numbers).

Answer: Necessary but not sufficient.

Method: Necessary: $a = 0 \Rightarrow ab = 0$ trivially. Not sufficient: $a = 1, b = 0$ gives $ab = 0$ but $a \neq 0$.

Investigate further: The correct sufficient-and-necessary condition for “ $a = 0$ ” from $ab = 0$ alone would need $b \neq 0$ added as an extra hypothesis.

4. Classify “ n^2 is even” as a condition for “ n is even”.

Answer: Necessary and sufficient.

Method: Sufficient: n even $\Rightarrow n^2$ even (Day 1, B2). Necessary: n^2 even $\Rightarrow n$ even, since if n were odd, n^2 would be odd (Day 1, B5, contrapositive-flavoured). Both directions hold.

Investigate further: Compare B1’s “wider/narrower” pattern: this pair is iff precisely because parity of n and parity of n^2 always agree, with no gap between them.

5. Classify “ x is an integer” as a condition for “ x is a rational number”.

Answer: Sufficient but not necessary.

Method: Sufficient: every integer is rational. Not necessary: $x = \frac{1}{2}$ is rational but not an integer.

Investigate further: Integers are a strict subset of the rationals, so “integer” can only ever be sufficient for “rational,” never necessary.

6. Classify “a triangle is equilateral” as a condition for “a triangle is isosceles”.

Answer: Sufficient but not necessary.

Method: Sufficient: an equilateral triangle has (at least) two equal sides, satisfying the definition of isosceles. Not necessary: a triangle with sides 3, 3, 5 is isosceles but not equilateral.

Investigate further: The same subset pattern as B5: equilateral triangles are a strict subset of isosceles triangles.

7. Given that “ $P \Rightarrow Q$ ” and “ $Q \Rightarrow R$ ” are both true, is P a sufficient condition for R ? Justify briefly.

Answer: Yes.

Method: If P holds, then $P \Rightarrow Q$ gives Q ; then $Q \Rightarrow R$ gives R . So $P \Rightarrow R$, i.e. P is sufficient for R .

Investigate further: This is the general transitivity-of-implication rule — Section C proves it abstractly (C5) rather than just applying it.

8. Given that “ n divisible by 9 $\Rightarrow$ n divisible by 3” and “ n divisible by 3 $\Rightarrow$ digit sum divisible by 3”, is “ n divisible by 9” sufficient for “digit sum divisible by 3”?

Answer: Yes.

Method: Chain the two implications: $9 \mid n \Rightarrow 3 \mid n \Rightarrow$ digit sum divisible by 3. By B7’s transitivity, $9 \mid n$ is sufficient for “digit sum divisible by 3.”

Investigate further: Chaining lets you skip straight from “divisible by 9” to a fact about digit sums without ever computing $n/3$ — exactly how speed is built from a toolkit.

9. Each of (a) n divisible by 8, (b) n divisible by 4, (c) n divisible by 2 is sufficient for “ n is even”. Which is the *weakest* (least restrictive) of the three?

Answer: (c) n divisible by 2.

Method: Every multiple of 8 is a multiple of 4, and every multiple of 4 is a multiple of 2, so (a) is the most restrictive and (c) the least. “ n divisible by 2” is in fact necessary and sufficient for “ n even” (they’re the same statement), making it the weakest condition that still works.

Investigate further: The weakest sufficient condition, when it exists, is exactly the necessary-and-sufficient one — anything weaker would stop guaranteeing the target.

10. Each of (a) n divisible by 2, (b) n divisible by 4, (c) n divisible by 8 is necessary for “ n is divisible by 24”. Which is the *strongest* (most restrictive) of the three?

Answer: (c) n divisible by 8.

Method: $24 = 8 \times 3$, and $8 \mid n \Rightarrow 4 \mid n \Rightarrow 2 \mid n$, so (c) implies both (a) and (b), making it the most restrictive (strongest) necessary condition among the three.

Investigate further: Among a chain of necessary conditions, the strongest one carries the most information — it gets you closest to the full truth (“divisible by 24”) without quite reaching it.

Section C Substitution & Structure

1. Which of the following best describes “ n is a multiple of 6” as a condition for “ n is a multiple of 2 and a multiple of 3”?

Answer: C) Necessary and sufficient.

Method: This restates B2 in the standard four-way TMUA classification format: both directions ($6 \mid n \Leftrightarrow 2 \mid n$ and $3 \mid n$) hold, as shown in B2.

Investigate further: This four-option layout (necessary-only / sufficient-only / both / neither) is the single most common TMUA Paper 2 question shape — you’ll see it again in Section D.

2. Prove that “ n is divisible by 4” is necessary and sufficient for “ n is even and $n/2$ is even”.

Answer: Proved both directions.

Method: Sufficient: if $4 \mid n$, write $n = 4k$; then n is even and $n/2 = 2k$ is even. Necessary: if n is even and $n/2$ is even, write $n/2 = 2k$, so $n = 4k$, i.e. $4 \mid n$.

Investigate further: This is a clean example of building an iff by proving each direction as its own short argument — the standard way to establish necessary-and-sufficient claims.

3. Classify “ $x^2 < 4$ ” as a condition for “ $x < 2$ ” (real x). Justify both directions.

Answer: Sufficient but not necessary.

Method: Sufficient: $x^2 < 4 \Leftrightarrow -2 < x < 2$, which implies $x < 2$. Not necessary: $x = -5$ satisfies $x < 2$ but $x^2 = 25 \not< 4$.

Investigate further: $x^2 < 4$ is actually necessary and sufficient for the two-sided condition $-2 < x < 2$ — restricting to the one-sided $x < 2$ is what breaks necessity.

4. Classify “ n is prime” as a condition for “ n has exactly two positive divisors”. Justify your answer.

Answer: Necessary and sufficient.

Method: This is the definition of a prime number, so it holds by construction: n is prime if and only if n has exactly two positive divisors (1 and n itself).

Investigate further: Definitions are always necessary-and-sufficient statements in disguise — that’s exactly what makes something a definition rather than a theorem.

5. Given that P is sufficient for Q , and Q is sufficient for R , prove that P is sufficient for R .

Answer: Proved.

Method: “ P sufficient for Q ” means $P \Rightarrow Q$; “ Q sufficient for R ” means $Q \Rightarrow R$. Assume P holds; then Q holds (first implication), then R holds (second implication). So $P \Rightarrow R$, i.e. P is sufficient for R .

Investigate further: This is B7/B8’s pattern proved in full generality, for any P, Q, R — once proved once, it can be reused instantly (as B8 did).

6. Is “ n is a multiple of 4” necessary for “ n is a multiple of 12”? Is it sufficient? Justify both answers.

Answer: Necessary: yes. Sufficient: no.

Method: Necessary: $12 = 4 \times 3$, so $12 \mid n \Rightarrow 4 \mid n$. Not sufficient: $n = 4$ is a multiple of 4 but not of 12.

Investigate further: This is a fully worked “necessary but not sufficient” classification, the mirror image of C1’s “necessary and sufficient” pair.

7. Find a condition C on a positive integer n that is necessary but not sufficient for “ n is a multiple of 100”. Justify your choice.

Answer: C : “ n is a multiple of 10.”

Method: Necessary: $100 = 10 \times 10$, so $100 \mid n \Rightarrow 10 \mid n$. Not sufficient: $n = 10$ is a multiple of 10 but not of 100.

Investigate further: Any divisor of 100 smaller than 100 itself (e.g. 4, 20, 25, 50) works by the same argument — necessity comes from being a factor, insufficiency from not being the whole thing.

8. Given that x is necessary and sufficient for y , and y is necessary and sufficient for z , prove that x is necessary and sufficient for z .

Answer: Proved.

Method: “ x necessary and sufficient for y ” gives both $x \Rightarrow y$ and $y \Rightarrow x$; likewise $y \Rightarrow z$ and $z \Rightarrow y$. Chaining sufficiency (C5): $x \Rightarrow y \Rightarrow z$ gives $x \Rightarrow z$. Chaining the reverse direction: $z \Rightarrow y \Rightarrow x$ gives $z \Rightarrow x$. Both directions hold, so x is necessary and sufficient for z .

Investigate further: “If” chains exactly like plain implication does, just in both directions at once — this is why a chain of definitions (as in C4) transfers equivalence all the way down the chain.

Section D Challenge

1. For real x , let P be “ $x > 0$ ” and Q be “ $x^2 > 0$ ”. Is P necessary and sufficient for Q , necessary but not sufficient, sufficient but not necessary, or neither?

Answer: C) Sufficient but not necessary.

Method: Sufficient: $x > 0 \Rightarrow x^2 > 0$ directly (a positive number squared is positive). Not necessary: $x = -1$ gives $x^2 = 1 > 0$, so Q holds, but P ($x > 0$) does not.

Investigate further: Compare Day 1’s A6: $x^2 > 0$ fails only at $x = 0$, so it holds for every nonzero real, positive or negative — far more than just $x > 0$ requires.

2. Classify “ x is rational” as a condition for “ x^2 is rational” (real x). Prove both parts of your classification.

Answer: Sufficient but not necessary.

Method: Sufficient: rationals are closed under multiplication, so $x = p/q$ rational gives $x^2 = p^2/q^2$ rational. Not necessary: $x = \sqrt{2}$ is irrational, yet $x^2 = 2$ is rational.

Investigate further: This is the reverse-engineered cousin of Day 1’s C3 (rational + irrational): squaring can destroy irrationality even though addition of a rational and an irrational never does.

3. A year Y is a leap year exactly when Y is divisible by 4, except that century years (multiples of 100) must be divisible by 400. Is “ Y is divisible by 4” necessary and sufficient, necessary but not sufficient, sufficient but not necessary, or neither, for “ Y is a leap year”? Justify with a specific year.

Answer: Necessary but not sufficient.

Method: Necessary: every leap year is divisible by 4 by definition. Not sufficient: $Y = 1900$ is divisible by 4 and by 100, but not by 400, so it is not a leap year.

Investigate further: The fully necessary-and-sufficient rule is: divisible by 4, and (not a century year, or divisible by 400) — the century-year clause is exactly what B3-style “extra hypothesis needed” patches.

4. Statements A, B, C satisfy $A \Rightarrow B$, $B \Rightarrow C$, and $C \Rightarrow A$. Prove that A, B, C are all logically equivalent (i.e. each pair is an “iff”).

Answer: Proved.

Method: We must show all six directions hold. $A \Rightarrow B$ and $B \Rightarrow C$ are given, and chaining them (C5)

gives $A \Rightarrow C$. Combined with the given $C \Rightarrow A$, this proves $A \Leftrightarrow C$. Similarly, $B \Rightarrow C \Rightarrow A$ (chaining) gives $B \Rightarrow A$; combined with the given $A \Rightarrow B$, this proves $A \Leftrightarrow B$. Finally $C \Rightarrow A \Rightarrow B$ gives $C \Rightarrow B$, which with $B \Rightarrow C$ proves $B \Leftrightarrow C$. All three pairs are iff.

Investigate further: A cycle of implications closing back on itself is a standard proof technique for establishing several equivalences at once without writing out all six directions individually — worth recognising on sight.

5. Statement S is: “If n is a multiple of 9, then n is a multiple of 3.” Is “ n is a multiple of 9” necessary and sufficient for “ n is a multiple of 3”? If not, precisely which of necessary/sufficient does it satisfy, and what does S ’s converse (from Day 1’s toolkit) have to do with your answer?

Answer: Sufficient but not necessary. S itself establishes sufficiency; S ’s converse (“if n is a multiple of 3, then n is a multiple of 9”) is false ($n = 6$), which is exactly why necessity fails.

Method: S is true ($9 \mid n \Rightarrow 3 \mid n$, since $9 = 3 \times 3$), giving sufficiency directly. Necessity would require S ’s converse to also be true, but $n = 6$ is a multiple of 3 and not of 9, so the converse is false, and necessity fails.

Investigate further: This is the general rule tying both days together: “ A necessary and sufficient for B ” is exactly “the statement $A \Rightarrow B$ and its converse are both true” — Day 1’s whole toolkit was secretly building toward Day 2’s vocabulary.